By hand calculations

20 december 2024

States

The state space is defined as:

$$O = \{(C,C)\}, \quad 1 = \{(D,C),(C,D),(D,D)\}$$

Payoff Matrix

The payoff matrix for the game is:

	C	D
C	$R = 0.8$	$S = 0$
D	$T = 1$	$P = 0.2$

Q-value Equations

First value of Q is action and the second is the current state. State 0:

1. Cooperation in 0:

$$Q_{c0} = (1 - \frac{\varepsilon}{2})\Big[R + \delta \max\{Q_{C0}, Q_{D0}\}\Big] + \frac{\varepsilon}{2}\Big[S + \delta \max\{Q_{C1}, Q_{D1}\}\Big]$$

2. Defection in 0:

$$Q_{D0} = (1 - \frac{\varepsilon}{2})\Big[T + \delta \max\{Q_{C1}, Q_{D1}\}\Big] + \frac{\varepsilon}{2}\Big[P + \delta \max\{Q_{C1}, Q_{D1}\}\Big]$$

State 1:

1. Cooperating in 1:

$$Q_{C1} = (1 - \frac{\varepsilon}{2})\Big[S + \delta \max\{Q_{C1}, Q_{D1}\}\Big] + \frac{\varepsilon}{2}\Big[R + \delta \max\{Q_{C0}, Q_{D0}\}\Big]$$

2. Defection in 1:

$$Q_{D1} = (1 - \frac{\varepsilon}{2})\Big[P + \delta \max\{Q_{C1}, Q_{D1}\}\Big] + \frac{\varepsilon}{2}\Big[T + \delta \max\{Q_{C1}, Q_{D1}\}\Big]$$

1 Scenarios

I will define the theoretical Q values of different strategies always against $\varepsilon - GT$

1.1 Case 1: Playing $\varepsilon - GT$

If the opponent plays $\varepsilon - GT$ and I play $\varepsilon - GT$, the Q-values are updated as:

$$Q_{C0} = (1-\frac{\varepsilon}{2})\Big[R+\delta Q_{C0}\Big] + \frac{\varepsilon}{2}\Big[S+\delta Q_{D1}\Big]$$

$$Q_{D0} = (1-\frac{\varepsilon}{2})\Big[T+\delta Q_{D1}\Big] + \frac{\varepsilon}{2}\Big[P+\delta Q_{D1}\Big]$$

$$Q_{C1} = (1-\frac{\varepsilon}{2})\Big[S+\delta Q_{D1}\Big] + \frac{\varepsilon}{2}\Big[R+\delta Q_{C0}\Big]$$

$$Q_{D1} = (1-\frac{\varepsilon}{2})\Big[P+\delta Q_{D1}\Big] + \frac{\varepsilon}{2}\Big[T+\delta Q_{D1}\Big]$$

Final System of Equations:

$$Q_{C0} = \frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{D1}}{1-(1-\frac{\varepsilon}{2})\delta}$$

$$Q_{D0} = (1-\frac{\varepsilon}{2})T + \frac{\varepsilon}{2}P + \delta Q_{D1}$$

$$Q_{C1} = (1-\frac{\varepsilon}{2})S + (1-\frac{\varepsilon}{2})\delta Q_{D1} + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{C0}$$

$$Q_{D1} = \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}$$

Substituting one in another: The reduced system of equations is given as follows:

$$Q_{C0} = \frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta \cdot \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}}{1-(1-\frac{\varepsilon}{2})\delta}$$

$$Q_{D0} = (1-\frac{\varepsilon}{2})T + \frac{\varepsilon}{2}P + \delta \cdot \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}$$

$$Q_{C1} = (1-\frac{\varepsilon}{2})S + (1-\frac{\varepsilon}{2})\delta \cdot \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta} + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{C0}$$

$$Q_{D1} = \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}$$

1.2 Case 2: Playing AD

If the opponent plays $\varepsilon - GT$ and I play AD, the Q-values are updated as :

$$Q_{C0} = (1-\frac{\varepsilon}{2})\Big[R+\delta Q_{D0}\Big] + \frac{\varepsilon}{2}\Big[S+\delta Q_{D1}\Big]$$

$$Q_{D0} = (1-\frac{\varepsilon}{2})\Big[S+\delta Q_{D1}\Big] + \frac{\varepsilon}{2}\Big[R+\delta Q_{D0}\Big]$$

$$Q_{C1} = (1-\frac{\varepsilon}{2})\Big[T+\delta Q_{D1}\Big] + \frac{\varepsilon}{2}\Big[P+\delta Q_{D1}\Big]$$

$$Q_{D1} = (1-\frac{\varepsilon}{2})\Big[P+\delta Q_{11}\Big] + \frac{\varepsilon}{2}\Big[T+\delta Q_{D1}\Big]$$

The reduced system of equations is given as follows:

$$Q_{D1} = \frac{\left(1-\frac{\varepsilon}{2}\right)P + \left(\frac{\varepsilon}{2}\right)T}{1-\delta} \tag{1}$$

$$Q_{D0} = \left(1-\frac{\varepsilon}{2}\right)T + \left(\frac{\varepsilon}{2}\right)P + \delta Q_{D1} \tag{2}$$

$$Q_{C1} = \left(1-\frac{\varepsilon}{2}\right)S + \left(1-\frac{\varepsilon}{2}\right)\delta Q_{D1} + \left(\frac{\varepsilon}{2}\right)R + \left(\frac{\varepsilon}{2}\right)\delta Q_{D0} \tag{3}$$

$$Q_{C0} = \left(1-\frac{\varepsilon}{2}\right)R + \left(1-\frac{\varepsilon}{2}\right)\delta Q_{D0} + \left(\frac{\varepsilon}{2}\right)S + \left(\frac{\varepsilon}{2}\right)\delta Q_{D1} \tag{4}$$

1.3 Case 3: Playing *AC*

$$Q_{00} = (1-\frac{\varepsilon}{2})\Big[R+\delta Q_{C0}\Big] + \frac{\varepsilon}{2}\Big[S+\delta Q_{D0}\Big]$$
$$Q_{01} = (1-\frac{\varepsilon}{2})\Big[T+\delta Q_{D0}\Big] + \frac{\varepsilon}{2}\Big[P+\delta Q_{D0}\Big]$$
$$Q_{10} = (1-\frac{\varepsilon}{2})\Big[S+\delta Q_{D0}\Big] + \frac{\varepsilon}{2}\Big[R+\delta Q_{C0}\Big]$$
$$Q_{11} = (1-\frac{\varepsilon}{2})\Big[P+\delta Q_{D0}\Big] + \frac{\varepsilon}{2}\Big[T+\delta Q_{D0}\Big]$$

The reduced system of equations is given as follows:

$$Q_{C1}^{\text{new}} = \frac{\left(1-\frac{\varepsilon}{2}\right)S + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{C0}}{1-\left(1-\frac{\varepsilon}{2}\right)\delta}$$

$$Q_{C0}^{\text{new}} = \frac{\left(1-\frac{\varepsilon}{2}\right)R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{C1}^{\text{new}}}{1-\left(1-\frac{\varepsilon}{2}\right)\delta}$$

This process repeats for a maximum of 100 iterations or until convergence is reached, i.e., when:

$$|Q_{C1}^{\text{new}} - Q_{C1}| < \text{tolerance} \quad \text{and} \quad |Q_{C0}^{\text{new}} - Q_{C0}| < \text{tolerance}$$

$$Q_{D1} = \left(1-\frac{\varepsilon}{2}\right)P + \frac{\varepsilon}{2}T + \delta Q_{C1}$$

$$Q_{D0} = \left(\left(1-\frac{\varepsilon}{2}\right)T + \frac{\varepsilon}{2}P\right) + \delta Q_{C1}$$